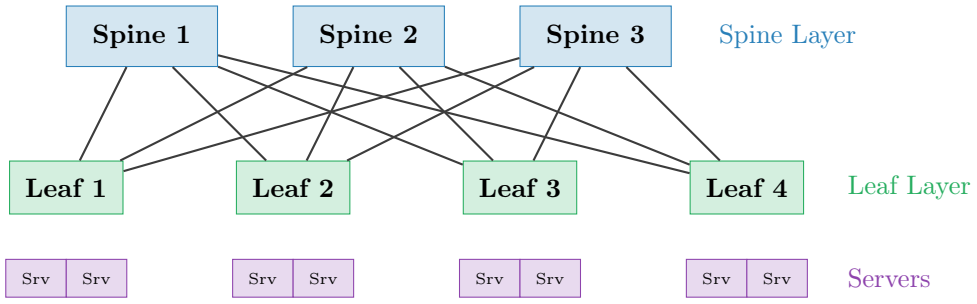
Leaf-Spine Topology

Complete Student Guide

Understanding Layer 3 Switches in
Modern Data Center Architectures



Topics Covered:

- Foundational Concepts
- Design & Architecture
- Routing Protocols
- Troubleshooting
- Comparisons & Trade-offs
- Real-World Implementation

A Comprehensive Q&A Reference

Version 2.0

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1 Foundational Questions

Q: If Layer 3 switches can route, why do we still need routers at all?

Layer 3 switches and routers serve different purposes despite both being capable of routing:

Layer 3 Switches Excel At:

- High-speed packet forwarding using hardware ASICs (Application-Specific Integrated Circuits)
- Routing between VLANs within a data center or campus
- Handling large volumes of internal (east-west) traffic
- Providing wire-speed performance with minimal latency

Routers Are Still Needed For:

- WAN connectivity (MPLS, internet edge, SD-WAN)
- Advanced services like NAT, VPN termination, deep packet inspection
- Complex policy-based routing and traffic engineering
- Protocol translation and connecting different network types
- Handling diverse interface types (serial, cellular, satellite)

In a typical leaf-spine data center, Layer 3 switches handle all internal routing, while dedicated routers sit at the edge for WAN and internet connectivity.

Q: What happens to traffic if a spine switch fails?

This is where the leaf-spine architecture demonstrates its resilience:

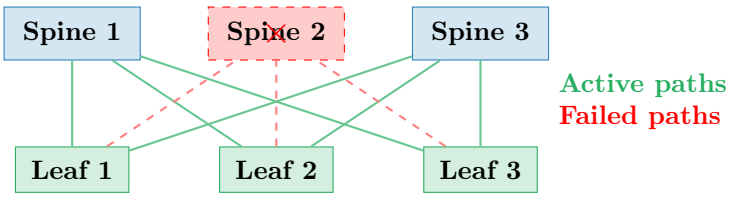


Figure 1: Spine failure scenario – traffic automatically reroutes through remaining spines

1. **Immediate Failover:** Every leaf switch is connected to every spine switch. If one spine fails, the leaf switches still have active paths through the remaining spines.
2. **ECMP Reconvergence:** Equal-Cost Multi-Path routing automatically redistributes traffic across the remaining spine switches. This happens within milliseconds when using protocols like BGP with BFD (Bidirectional Forwarding Detection).
3. **Capacity Reduction:** The network continues to function but with reduced aggregate bandwidth. For example, if you have 4 spines and lose 1, you lose 25% of your spine capacity.

4. **No Single Point of Failure:** Unlike traditional three-tier designs with core switch dependencies, leaf-spine ensures no single spine failure isolates any leaf.

Design Recommendation

Always deploy at least 2 spine switches (preferably 4+) and ensure each leaf has sufficient uplink bandwidth to handle spine failures.

Q: Why is it called 'leaf-spine' and not just 'two-tier'?

The terminology comes from the biological analogy and describes the functional roles:

Spine:

- Acts as the backbone of the network
- Provides the high-speed interconnection fabric
- Never connects directly to endpoints—only to leaf switches
- Like a human spine, it's the central support structure

Leaf:

- The edge layer where endpoints (servers, storage) connect
- "Leaves" branch out from the spine to reach endpoints
- Each leaf connects to every spine (full mesh to spine layer)

While it is technically a two-tier architecture, "leaf-spine" specifically implies:

- Full mesh connectivity between tiers
- Layer 3 routing at both tiers
- ECMP-based load balancing
- No leaf-to-leaf direct connections

A generic "two-tier" design could be Layer 2, could use spanning tree, and might not have full mesh connectivity. The term "leaf-spine" carries specific architectural implications.

Q: What's the difference between east-west and north-south traffic?

These terms describe traffic flow direction in data center networks:

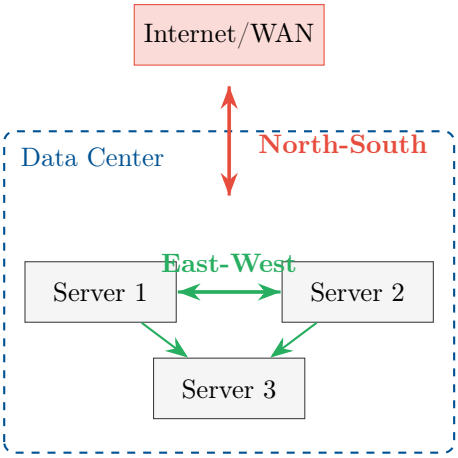


Figure 2: Traffic flow directions in a data center

Aspect	East-West Traffic	North-South Traffic
Direction	Lateral, between servers	Vertical, in/out of data center
Examples	Server-to-server, VM-to-VM, database replication	Client requests from internet, backup to cloud
Volume	70–80% of modern DC traffic	20–30% of modern DC traffic
Optimized By	Leaf-spine architecture	Edge routers, firewalls, load balancers
Latency Needs	Ultra-low (microseconds)	More tolerant (milliseconds acceptable)

Table 1: Comparison of east-west and north-south traffic characteristics

Traditional three-tier networks were optimized for north-south traffic (clients outside accessing servers inside). Modern applications (microservices, distributed databases, big data) generate massive east-west traffic, which is why leaf-spine became the preferred architecture.

2 Design & Architecture

Q: How do I decide how many spine switches I need?

The number of spine switches depends on several factors:

1. Port Density Requirements:

- Each spine must have enough ports to connect to every leaf
- Formula: Minimum spine ports $\geq$ Number of leaf switches
- Example: 48 leaves require spines with at least 48 ports each

2. Bandwidth Requirements:

- Calculate total required east-west bandwidth
- Each spine adds aggregate bandwidth capacity
- More spines = more parallel paths = more bandwidth

3. Redundancy Goals:

- Minimum: 2 spines (single failure tolerance)
- Recommended: 4 spines (allows maintenance + failure)
- High availability: 4+ spines with N+2 redundancy

4. Oversubscription Ratio:

- Calculate: (Total leaf downlink bandwidth) / (Total leaf uplink bandwidth)
- Lower oversubscription = more spines needed
- 3:1 oversubscription is common; 1:1 for high-performance workloads

Practical Example

- 32 leaf switches, each with 48× 10G server ports (480 Gbps downlink)
- Target oversubscription: 3:1
- Required uplink per leaf: 160 Gbps (4× 40G or 4× 100G)
- With 4× 100G uplinks per leaf → **4 spine switches needed**

Q: Can leaves connect to each other, or only to spines?

Standard Rule

Leaves should NOT connect directly to each other.

This is a fundamental principle of leaf-spine design for several reasons:
Why No Leaf-to-Leaf Links:

- Violates the predictable traffic pattern (always exactly 2 hops)
- Complicates routing and can create suboptimal paths
- Makes capacity planning unpredictable
- Can lead to traffic engineering challenges

Exceptions (Rare):

- vPC/MLAG peer links between leaf switch pairs (for Layer 2 redundancy)
- Temporary migration scenarios
- Specific vendor implementations like Cisco ACI's leaf pairs

The Two-Hop Guarantee

In a proper leaf-spine design, traffic between any two endpoints traverses exactly:

1. Source leaf
2. One spine (chosen by ECMP)
3. Destination leaf

This predictability is a key advantage of the architecture.

Q: Where do I connect my WAN router—to a leaf or a spine?

Best Practice: Connect WAN routers to dedicated "border leaf" switches.

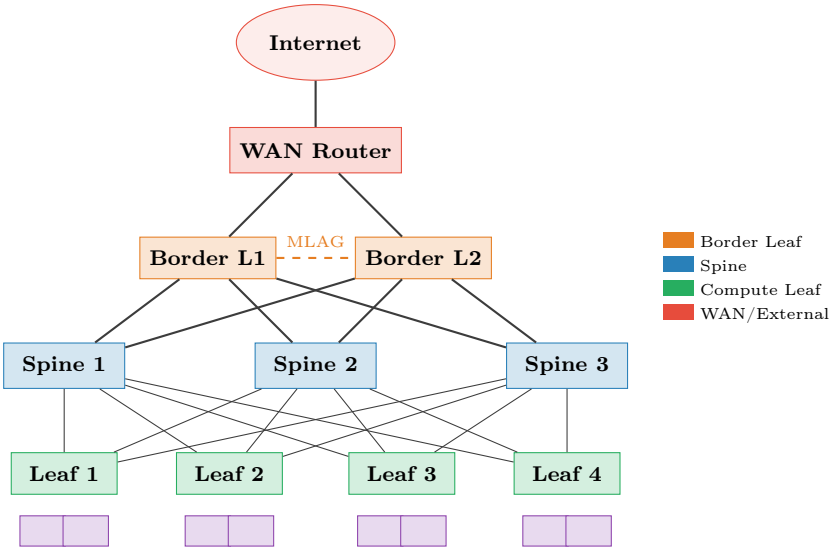


Figure 3: Border leaf topology for WAN connectivity

Why Border Leaves (Not Spines):

- Maintains the architectural principle that spines only connect to leaves
- Provides a clear demarcation point for external connectivity
- Allows for dedicated security policies and QoS at the border
- Scales independently—add more border leaves for more WAN capacity
- Keeps spine switches focused on internal fabric performance

Border Leaf Functions:

- Connect to WAN routers, firewalls, load balancers
- Handle north-south traffic aggregation
- Apply external-facing policies
- Peer with external BGP neighbors

Q: How do I handle servers that need to be in the same VLAN across multiple racks?

This is one of the most common challenges in leaf-spine designs. There are several approaches:

1. VXLAN (Preferred Solution):

- Encapsulates Layer 2 frames in UDP packets
- Extends VLANs across the Layer 3 fabric as "overlay" networks
- Each VXLAN segment identified by a 24-bit VNI (VXLAN Network Identifier)
- Supports up to 16 million logical networks (vs. 4094 VLANs)
- Leaf switches act as VXLAN Tunnel Endpoints (VTEPs)

2. EVPN-VXLAN (Enterprise Standard):

- Combines VXLAN with EVPN control plane
- Uses BGP to distribute MAC/IP information

- Provides optimal forwarding and multi-tenancy
- Eliminates flood-and-learn behavior

3. Design Around It:

- Keep broadcast domains within a single leaf or leaf pair
- Use Layer 3 everywhere with /31 or /32 host routes
- Deploy applications that don't require Layer 2 adjacency

What NOT To Do

- Don't stretch VLANs using spanning tree across the fabric
- Don't create Layer 2 links between leaves
- Don't disable routing on spines to create a large Layer 2 domain

Q: What's the oversubscription ratio and why does it matter?

Definition: Oversubscription ratio compares the total bandwidth of endpoints (downlinks) to the total bandwidth available to the fabric (uplinks).

$$OversubscriptionRatio = \frac{TotalDownlinkBandwidth}{TotalUplinkBandwidth}$$

Example Calculation:

- Leaf switch: 48 × 10G server ports = 480 Gbps downlink
- Leaf switch: 6 × 40G uplinks to spines = 240 Gbps uplink
- Oversubscription: 480/240 = **2:1**

Ratio	Use Case	Cost
1:1	High-frequency trading, HPC, storage	Highest
2:1	Database clusters, real-time analytics	High
3:1	General compute, virtualization	Moderate
4:1+	Dev/test, low-utilization workloads	Lower

Table 2: Common oversubscription ratios and their applications

Why It Matters:

- Higher oversubscription = potential congestion during peak usage
- Lower oversubscription = more consistent performance but higher cost
- Must match workload requirements—storage traffic typically needs lower ratios
- Affects spine switch quantity and uplink port speeds

3 Protocol-Related Questions

Q: Why use BGP in a data center instead of OSPF?

While OSPF can work in leaf-spine deployments, BGP has become the de facto standard for several reasons:

BGP Advantages in Data Centers:

- **Scalability:** BGP handles large routing tables efficiently; OSPF's link-state database becomes unwieldy at scale
- **Stability:** BGP's path-vector nature prevents routing loops inherently; no complex area design needed
- **Granular Control:** Extensive policy capabilities for traffic engineering
- **Multipath:** BGP multipath/ECMP works well with leaf-spine
- **Convergence:** With BFD, BGP converges in sub-second timeframes
- **Simplicity:** eBGP (using private ASNs) eliminates the need for complex OSPF area design
- **EVPN Support:** BGP is the control plane for EVPN-VXLAN overlays

Common BGP Design for Leaf-Spine

- Each leaf: unique private ASN (e.g., 65001, 65002, ...)
- All spines: single shared ASN (e.g., 65000)
- eBGP peering between every leaf and every spine
- Multipath enabled for ECMP across all spines

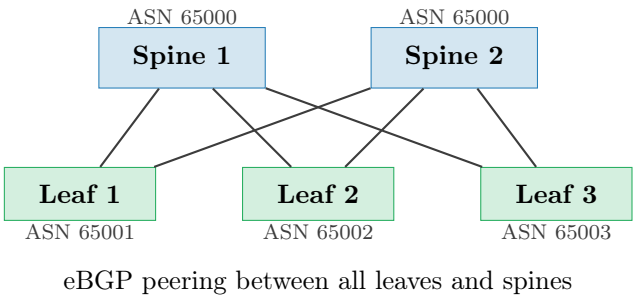


Figure 4: BGP ASN assignment in leaf-spine topology

When OSPF Might Still Work:

- Small deployments (under 50 switches)
- Teams with strong OSPF expertise but limited BGP experience
- Integration with existing OSPF campus networks

Q: How does ECMP actually decide which path to take?

ECMP (Equal-Cost Multi-Path) distributes traffic across multiple equal-cost paths using a hashing algorithm:

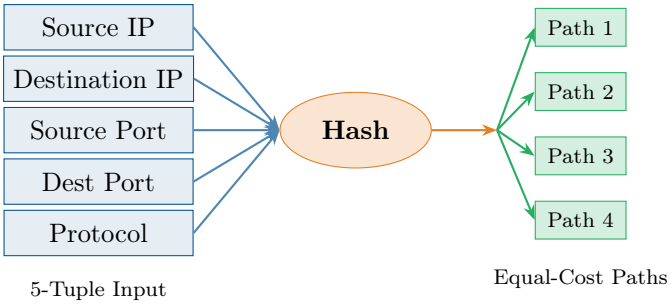


Figure 5: ECMP hash-based path selection

How It Works:

1. Switch calculates a hash value from the packet header fields (5-tuple)
2. Hash result is used to select one of the available equal-cost next-hops
3. All packets with the same hash (same flow) follow the same path
4. Different flows get distributed across paths

Key Characteristics:

- **Per-flow, not per-packet:** Maintains packet ordering within a flow
- **Deterministic:** Same 5-tuple always selects the same path
- **Stateless:** Each switch makes independent decisions

Potential Issues:

- **Elephant flows:** Large, long-lived flows can create imbalance
- **Polarization:** Symmetric hashing at multiple layers can concentrate traffic
- **Solutions:** Flowlet switching, asymmetric hashing, adaptive routing

Q: What is VXLAN and when do I need it?

VXLAN = Virtual Extensible LAN

VXLAN is a network virtualization technology that encapsulates Layer 2 Ethernet frames within Layer 3 UDP packets.

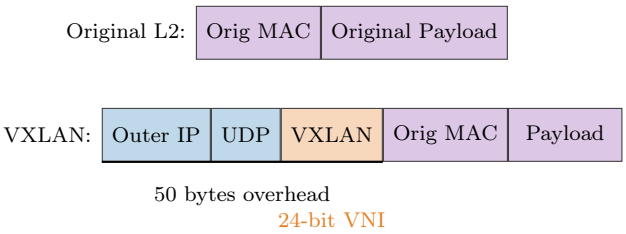


Figure 6: VXLAN encapsulation format

Technical Details:

- Encapsulation: Original L2 frame + VXLAN header + UDP header + IP header
- Standard port: UDP 4789
- Identifier: 24-bit VNI (VXLAN Network Identifier) = 16+ million segments
- MTU consideration: Adds 50 bytes overhead; requires jumbo frames or reduced MTU

When You Need VXLAN:

- Extending Layer 2 domains across a Layer 3 fabric
- Multi-tenancy with overlapping IP addresses
- VM mobility across racks without IP changes
- Scaling beyond 4094 VLANs
- Network microsegmentation
- Hybrid cloud connectivity

When You Don't Need VXLAN:

- Pure Layer 3 design with no Layer 2 extension requirements
- Small deployments where VLANs within leaf pairs suffice
- Environments where applications don't require Layer 2 adjacency

VXLAN Control Planes:

- **Flood-and-learn:** Simpler but less efficient
- **EVPN:** BGP-based, eliminates flooding, industry standard
- **SDN controller:** Centralized control (VMware NSX, Cisco ACI)

Q: How do leaf switches know where a destination host is?

This depends on whether the destination is in the same subnet (Layer 2) or different subnet (Layer 3):

Same Subnet (Layer 2 Lookup):

1. Leaf maintains a MAC address table
2. Learns MAC addresses from incoming frames (source MAC + ingress port)
3. Forwards based on destination MAC
4. Unknown unicast: floods or uses overlay control plane

Different Subnet (Layer 3 Lookup):

1. Host sends packet to default gateway (leaf switch)
2. Leaf performs routing table lookup for destination IP
3. Determines next-hop (spine switch) via ECMP
4. Destination leaf performs final Layer 2 lookup

With EVPN-VXLAN:

1. Leaf switches advertise local MAC/IP bindings via BGP EVPN
2. All leaves learn remote host locations from BGP updates
3. No flooding needed—destination VTEP is known from control plane
4. Provides both Layer 2 and Layer 3 reachability information

Key Tables on a Leaf Switch

- **MAC address table:** MAC → port mapping
- **ARP table:** IP → MAC mapping
- **Routing table:** IP prefix → next-hop
- **EVPN table:** MAC/IP → remote VTEP

Q: What's the difference between underlay and overlay networks?

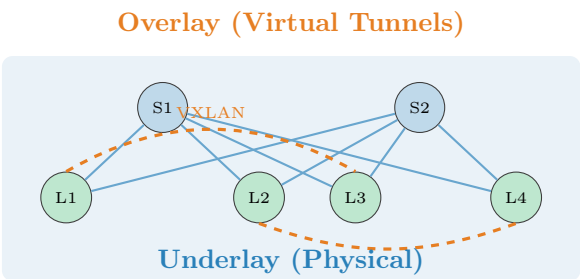


Figure 7: Underlay vs. overlay network relationship

Aspect	Underlay	Overlay
Purpose	Physical connectivity	Logical segmentation
Protocol	BGP/OSPF for IP routing	EVPN-VXLAN for L2/L3 VPNs
Addressing	Loopback IPs, P2P links	VNIs, tenant IP spaces
Visibility	Infrastructure team	Tenant/application team
Concerns	Physical topology, bandwidth, latency	Logical topology, segmentation, policy

Table 3: Underlay vs. overlay comparison

Analogy: Think of it like the postal system. The underlay is the physical infrastructure (roads, trucks, sorting facilities). The overlay is the logical addressing system (addresses, ZIP codes, routes) that determines how mail is organized and delivered.

4 Troubleshooting & Operations

Q: How do I trace a packet's path when there are multiple equal-cost routes?

Traditional traceroute often shows inconsistent results with ECMP because each probe may take a different path. Here are better approaches:

1. Paris Traceroute:

- Modifies probe packets to ensure same hash = same path
- Keeps flow identifier consistent across probes
- Available on Linux: `paris-traceroute`

2. Vendor-Specific Tools:

- Cisco: `traceroute` with specific source/destination ports
- Arista: `show ip route` with `detail` to see ECMP paths
- Cumulus: `mtr` or `tracepath`

3. Switch-Based Verification:

```
# Check which next-hop a specific flow will use:
show ip route <destination> | include ECMP
show forwarding route <destination> detail

# Verify ECMP hash result:
```

```
show hardware forwarding ecmp lookup <src-ip> <dst-ip> <src-port>
   <dst-port> <protocol>
```

4. Packet Capture Analysis:

- Capture on multiple paths simultaneously
- Use SPAN/mirror sessions on potential paths
- Correlate by packet sequence numbers

5. Network Telemetry:

- Modern switches support INT (In-band Network Telemetry)
- Packets carry path information through the fabric
- Analytics platforms visualize actual paths

Q: What happens if one leaf-to-spine link is slower than the others?

This is a significant problem because ECMP assumes equal bandwidth across all paths:

Symptoms:

- Unpredictable latency (some flows experience delays)
- Congestion on the slower link while others are underutilized
- Packet drops when slower link buffer fills
- Inconsistent application performance

Why ECMP Doesn't Handle This:

- ECMP distributes flows equally, ignoring bandwidth differences
- Hash-based selection doesn't consider link utilization
- A 10G link and 100G link would receive equal flow distribution

Solutions:

1. **Fix the Root Cause:** Replace the faulty link/transceiver. All links should be identical speed.
2. **Remove the Slow Link from ECMP:** Adjust routing metrics to exclude it or administratively shut down until fixed.
3. **Use Weighted ECMP (UCMP):** Unequal-Cost Multi-Path distributes based on bandwidth. Not all platforms/protocols support this. BGP: use DMZ bandwidth community.
4. **Traffic Engineering:** Segment routing to steer traffic explicitly or SDN controller-based path selection.

Best Practice

Design with homogeneous links. All leaf-to-spine connections should be identical speed. Monitor for interface errors and speed mismatches.

Q: How quickly does the network reconverge after a failure?

Reconvergence time depends on detection mechanism and routing protocol:

Detection Method	Detection Time	Notes
Physical link down	10–50 ms	Immediate for direct failures
BFD (aggressive)	50–150 ms	50ms × 3 = 150ms typical
BFD (standard)	300–900 ms	300ms × 3
BGP keepalives	90–180 seconds	Default hold time
OSPF hello timers	40 seconds	Default dead interval

Table 4: Failure detection times by method

Routing Protocol Convergence:

- BGP with BFD: Sub-second (typically 200–500ms total)
- OSPF with BFD: Similar to BGP when tuned
- Without BFD: Several seconds to minutes

Best Practice Configuration for Fast Convergence:

```
! BGP example
router bgp 65001
  neighbor <spine-ip> bfd

! BFD configuration (300ms detection time)
bfd interval 100 min-rx 100 multiplier 3
```

Total Failover Timeline

1. Failure occurs: 0 ms
2. BFD detection: ∼150 ms
3. BGP path withdrawal: ∼50 ms
4. ECMP recalculation: ∼10 ms
5. Hardware programming: ∼50 ms
6. **Total: approximately 200–300 ms**

Q: How do I troubleshoot asymmetric routing issues?

Asymmetric routing (traffic takes different paths in each direction) is normal and expected in leaf-spine networks. It only becomes a problem in specific situations:

When Asymmetry Is a Problem:

- Stateful firewalls that see only one direction
- Network monitoring tools expecting symmetric flows
- Certain load balancer health checks
- QoS policies applied unidirectionally

Troubleshooting Steps:

1. Identify the asymmetry:

```
# Trace forward path
traceroute <destination>

# Trace reverse path (from destination device)
traceroute <source>
```

```
# Compare the paths
```

2. **Verify routing tables:**

```
show ip route <destination>
show ip bgp <destination>
# Check for unequal metrics or missing paths
```

3. **Check for policy-based routing:**

```
show route-map
show ip policy
```

Solutions:

- **Stateful devices:** Deploy in active-active pairs with session sync
- **Firewalls:** Use clustering or distributed firewall designs
- **Load balancers:** Use Direct Server Return (DSR) or SNAT
- **Monitoring:** Capture at multiple points and correlate

Key Insight

In a properly designed leaf-spine network, asymmetric routing is normal. Design your services and security to accommodate it rather than trying to force symmetry.

5 Comparison & Trade-off Questions

Q: Why not just use Spanning Tree with Layer 2 switches—isn't that simpler?

While Spanning Tree Protocol (STP) based designs are conceptually simpler, they have significant limitations that make them unsuitable for modern data centers:

Aspect	Spanning Tree (Layer 2)	Leaf-Spine (Layer 3)
Active paths	1 (others blocked)	All paths active (ECMP)
Bandwidth utilization	~50% typical	Near 100%
Convergence	30–50 seconds (STP)	Sub-second (BGP+BFD)
Scale	100s of switches	1000s of switches
Failure domain	Entire VLAN	Single link/switch
Loop prevention	Blocking ports	Layer 3 routing

Table 5: Spanning Tree vs. Leaf-Spine comparison

Why STP Falls Short:

- **Wasted capacity:** Blocked links sit idle during normal operation
- **Slow convergence:** STP takes 30–50 seconds; RSTP is faster but still seconds
- **Broadcast storms:** Misconfiguration can bring down entire Layer 2 domain

- **Limited scale:** Large Layer 2 domains are fragile and hard to troubleshoot
- **East-west inefficiency:** Traffic may traverse suboptimal paths

When Layer 2 With STP Might Be Acceptable:

- Very small environments (<10 switches)
- Legacy applications requiring Layer 2 adjacency with no budget for VXLAN
- Temporary/lab environments

Bottom Line

The initial simplicity of STP is outweighed by operational complexity, performance limitations, and fragility at scale.

Q: When would I choose a traditional three-tier design over leaf-spine?

Three-tier (access-distribution-core) designs still make sense in certain scenarios:

Choose Three-Tier When:

- **Campus networks:** Geographic distribution across buildings favors hierarchical design
- **Predominantly north-south traffic:** Traditional client-server applications
- **Existing infrastructure:** Large investment in three-tier equipment
- **WAN aggregation:** When WAN optimization and policy are centralized
- **Small scale:** Under 500 endpoints with simple requirements
- **Team expertise:** Staff deeply familiar with hierarchical designs

Choose Leaf-Spine When:

- Data center or cloud environment
- Heavy east-west traffic (microservices, distributed apps)
- Need for horizontal scaling
- Predictable latency requirements
- Multi-tenancy or network virtualization needs
- Modern automation and CI/CD integration

Hybrid Approach: Many organizations use both: leaf-spine in the data center, three-tier in the campus, connected via border leaves and WAN routers.

Q: Is leaf-spine only for data centers, or can I use it in a campus network?

Leaf-spine can be used in campus networks, though it's less common. Here's the analysis:

Campus Leaf-Spine Advantages:

- Consistent architecture across data center and campus
- Better east-west performance for modern campus applications
- Simplified operations with uniform design
- Good for wireless-first campuses with distributed APs

Campus Leaf-Spine Challenges:

- **Geography:** Campus buildings are spread out; running fiber to centralized spines can be costly
- **Cost:** Layer 3 switches at every closet are more expensive than Layer 2
- **Legacy support:** Older devices may require Layer 2 adjacency

- **Power/space:** IDF closets may not accommodate Layer 3 switches

Campus Leaf-Spine Architectures:

1. **Centralized:** Spines in main data closet, fiber to leaves in each IDF. Works for single-building campus.
2. **Distributed:** Spine layer in each building, super-spine connecting buildings ("leaf-spine-super-spine" hierarchy).
3. **SD-Access (Cisco Example):** Fabric-based campus with VXLAN overlay, leaf-spine principles with campus-specific features, centralized policy and automation.

Recommendation

For new campus builds, consider leaf-spine if you have centralized closets and modern applications. For distributed campuses with many IDFs, traditional three-tier or SD-Access may be more practical.

Q: What's the cost difference between Layer 2 and Layer 3 leaf-spine?

The cost comparison involves multiple factors:

Component	Layer 2 Design	Layer 3 Leaf-Spine
Switch cost	Lower (L2 ASICs)	Higher (L3 routing capability)
Uplink ports	Fewer (blocked by STP)	More (all paths active)
Transceivers	Fewer	More (full mesh)
Cabling	Fewer cables	More cables

Table 6: Hardware cost comparison

Approximate Price Difference:

- Layer 3 switches: 10–30% more than equivalent Layer 2
- Additional cabling/transceivers for full mesh: 20–40% more
- Overall CAPEX: 15–40% higher for leaf-spine

Operational Cost Factors:

Layer 2 Hidden Costs:

- More time troubleshooting STP issues
- Longer outages during convergence
- Higher risk of broadcast storms
- Limited scalability requiring redesign sooner

Layer 3 Leaf-Spine Savings:

- Faster troubleshooting with Layer 3 tools
- Better automation and programmability
- Longer architecture lifespan
- More efficient bandwidth utilization

TCO Perspective

Over a 5-year lifecycle, leaf-spine often has **lower TCO** despite higher initial costs because of better performance (fewer switches needed), simpler operations (lower staffing costs), higher reliability (less downtime cost), and easier scaling (delayed forklift upgrades).

6 Practical & Real-World Questions

Q: What vendors are commonly used for leaf-spine deployments?

The leaf-spine market includes traditional networking vendors and newer entrants:

Vendor	Key Products	Notes
Cisco	Nexus 9000 series	ACI for SDN; NX-OS or ACI mode
Arista	7050/7060 (leaf), 7500 (spine)	Strong in cloud/finance; Linux-based EOS
Juniper	QFX5000 series	Apstra for fabric management
NVIDIA	Spectrum series	Popular for AI/ML; Cumulus Linux OS
Dell	PowerSwitch S-series	OS10; cost-effective option
White Box	Edgecore, Delta, Celestica	SONiC, Cumulus, DENT OS; lowest cost

Table 7: Major vendors for leaf-spine deployments

Selection Criteria:

- Existing vendor relationships and expertise
- Required features (EVPN, specific protocols)
- Automation and API capabilities
- Support and professional services
- Total cost of ownership

Q: How does this work with VMware/Kubernetes/cloud environments?

Modern virtualization and container platforms integrate with leaf-spine fabrics:

VMware Integration:

- **vSphere Standard:** VLANs mapped to port groups; fabric provides underlay
- **NSX-T:** Overlay networking with VXLAN/Geneve; fabric is pure underlay
- **vDS (Distributed Switch):** Logical switching across hosts; fabric transports
- **Key point:** Leaf-spine provides reliable, fast underlay; NSX handles overlay

Kubernetes Integration:

- **CNI (Container Network Interface):** Calico, Cilium, Flannel handle pod networking
- **BGP peering:** Some CNIs (Calico) peer directly with leaf switches
- **LoadBalancer services:** MetalLB can integrate with fabric BGP
- **Service mesh:** Istio/Linkerd operate at Layer 7; fabric handles Layer 3

Public Cloud Connectivity:

- **Direct Connect/ExpressRoute:** Terminate on border leaves
- **VPN tunnels:** Terminate on dedicated appliances or border leaves
- **Hybrid overlay:** Extend VXLAN fabrics to cloud VPCs

Best Practices

- Enable jumbo frames (MTU 9000+) for VXLAN overhead
- Configure ECMP for east-west distribution
- Use BGP peering where possible for dynamic routing
- Ensure consistent QoS marking across physical and virtual

Q: Do I need the same model switch for all leaves? All spines?

Short Answer: No, but homogeneity simplifies operations.

Leaf Switch Considerations:

Can differ in:

- Port count (24 vs. 48 ports)
- Port speed (10G vs. 25G access ports)
- Features (GPU compute leaves vs. storage leaves)

Should be consistent:

- Uplink speed to spines (all 100G, for example)
- Operating system version
- Control plane capabilities (BGP, EVPN support)

Spine Switch Considerations:

Strongly Recommended: Identical Spines

- Same model and port configuration
- Same line card types if modular
- Same software version

Why: ECMP assumes equal-cost paths; capacity planning is simpler; troubleshooting is easier; spare parts inventory is reduced.

Common Mixed Scenarios:

1. Different leaf types (acceptable):

- Compute leaves: 48× 25G + 8× 100G
- Storage leaves: 24× 100G + 8× 100G
- Border leaves: 48× 10G + 8× 100G + 4× 10G WAN

2. Generational differences (use caution):

- Older leaves with 40G uplinks mixed with newer leaves with 100G uplinks
- Requires careful capacity planning

Recommendation

Standardize on 2–3 leaf SKUs maximum. Keep all spines identical. Document the role of each switch type. Maintain consistent OS versions across the fabric.

Q: How do I migrate from a legacy three-tier network to leaf-spine?

Migration requires careful planning to minimize disruption. Here's a phased approach:

Phase 1: Assessment and Planning

- Inventory current network (devices, VLANs, traffic flows)

- Identify dependencies and application requirements
- Design target leaf-spine architecture
- Create IP addressing and VLAN mapping plan
- Define success criteria and rollback procedures

Phase 2: Parallel Build

- Deploy leaf-spine fabric alongside existing network
- Configure routing, VXLAN, and policies
- Test thoroughly with non-production workloads
- Validate performance and failover

Phase 3: Interconnection

- Connect legacy core to border leaves
- Establish routing between old and new fabrics
- Extend VLANs if needed (VXLAN gateway to legacy VLAN)
- Enable traffic flow between fabrics

Phase 4: Workload Migration

- Migrate workloads in waves (lowest risk first)
- Move by rack, application, or VLAN depending on dependencies
- Options: Physical move (recable servers), VM migration (vMotion), IP renumbering
- Validate each wave before proceeding

Phase 5: Legacy Decommission

- Remove dependencies on old fabric
- Decommission legacy switches
- Update documentation and monitoring

Strategy	Description	Risk	Timeline
Big Bang	Weekend cutover; all at once	Higher	Fastest
Phased	Migrate rack-by-rack or app-by-app	Lower	Longer
Greenfield Island	New apps to leaf-spine; legacy stays	Lowest	Longest

Table 8: Migration strategy comparison

Key Success Factors

- Detailed network discovery before starting
- Application team involvement
- Clear rollback plans for each phase
- Monitoring during and after migration
- Patience—rushed migrations cause outages

7 Quick Reference Tables

7.1 Leaf-Spine vs. Three-Tier Comparison

Feature	Three-Tier	Leaf-Spine
Topology	Hierarchical (Access-Dist-Core)	Flat (Leaf-Spine)
Blocking	Oversubscribed at distribution	Non-blocking capable
Path diversity	Limited by STP	Full ECMP
Scalability	Vertical (bigger switches)	Horizontal (more leaves)
Latency	Variable (3–5 hops)	Predictable (2 hops)
Best for	Campus, north-south traffic	Data center, east-west traffic

7.2 Protocol Comparison for Leaf-Spine

Protocol	Use Case	Pros	Cons
eBGP	Underlay routing	Scalable, flexible, EVPN-ready	More complex config
OSPF	Underlay (small scale)	Simpler, familiar	Area design complexity
EVPN	Overlay control plane	Optimal forwarding, multi-tenant	Learning curve
VXLAN	L2 over L3 extension	16M segments, standard	MTU overhead

7.3 Common Leaf-Spine Switch Recommendations

Role	Typical Specs	Example Models
Compute Leaf	48× 10/25G + 4–8× 100G	Arista 7050X3, Cisco N9K-C93180YC
Storage Leaf	24–32× 100G + 8× 100/400G	Arista 7060X4, Cisco N9K-C9364C
Spine	32–64× 100G or 400G	Arista 7500R3, Cisco N9K-C9508
Border Leaf	Mixed: 10G + 100G + WAN	Similar to compute + WAN ports

7.4 Convergence Time Reference

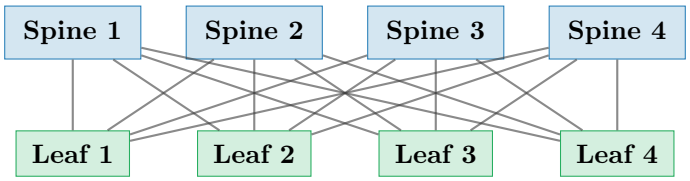
Technology	Typical Time	Notes
STP	30–50 seconds	Classic spanning tree
RSTP	1–5 seconds	Rapid spanning tree
OSPF (default)	40 seconds	Default dead interval
OSPF + BFD	200–500 ms	With aggressive BFD
BGP (default)	90–180 seconds	Default hold time
BGP + BFD	200–300 ms	Sub-second with BFD

Conclusion

Leaf-spine topology has become the de facto standard for modern data center networks due to its scalability, predictable performance, and support for east-west traffic patterns. Key takeaways:

- Every leaf connects to every spine, providing full redundancy and ECMP
- BGP is the preferred routing protocol for its scalability and EVPN support

- VXLAN with EVPN enables Layer 2 extension across the Layer 3 fabric
- Proper design accounts for oversubscription, failure domains, and growth
- Migration from legacy networks should be phased and well-planned



Full mesh: Every leaf connects to every spine

Understanding these concepts prepares students for real-world data center design and operations roles.

End of Document

Version 2.0 — Network Architecture Reference